Materials Bridge Line B: Magnon Address Stability in alpha-Fe_2O_3 for HAOS-IIP

Tomislav Rupić
Independent Researcher

May 2026

Abstract

This numbered update records release `65.1`: Materials Bridge Line B for HAOS-IIP. Line B is a bounded spin-sector materials bridge around coherent sub-THz magnon propagation in alpha-Fe_2O_3. It is a first-class repository artifact, but it is not yet a locally reproduced raw-data bridge. The current sidecar records a literature/user-audit-seeded persistence proxy of `0.902`, the reported group-velocity range `17.31--24.43 km/s`, micrometer-scale directional propagation, and the explicit raw-data boundary `NO_PUBLIC_STFT_GRID_FOUND`. The main literature anchor is Li et al., "Anisotropic Coherent Propagation of Sub-Terahertz Magnons in Altermagnetic alpha-Fe_2O_3," DOI `10.1002/adom.202503604`. The secondary alpha-Fe_2O_3 control context is Yang et al., "Spin-orbit torque manipulation of sub-terahertz magnons in antiferromagnetic alpha-Fe_2O_3," DOI `10.1038/s41467-024-48431-w`. This release does not claim a proof of HAOS-IIP, does not introduce a new magnon theory, and does not claim a $k_\star$ result without raw sweep data. Its function is bookkeeping and boundary-setting: the alpha-Fe_2O_3 magnon address-stability prediction layer is now represented by an auditable sidecar artifact, while raw-data validation remains pending.

Contents

1 Status and Scope

This paper is release `65.1` in the HAOS-IIP numbered paper sequence. It records:

- a new external spin-sector materials sidecar;
- a first-class alpha-Fe_2O_3 magnon bridge artifact;
- a bounded persistence summary seeded by a user-provided audit anchor and literature checks;
- public-source references and explicit evidence class;
- a raw-data boundary that prevents accidental overclaiming;
- a link to the four magnon prediction records `HAOS-PRED-010` through `HAOS-PRED-013`.

The sidecar path is:

```
experiments/materials_bridge/magnon_alpha_fe2o3_audit/
```

The release does not modify the HAOS-IIP frozen core, frozen phase artifacts, telemetry definitions, theory files, prior experiments, or prior paper releases.

2 Strict Non-Claims

This release does not claim:

- that alpha-Fe_2O_3 proves HAOS-IIP;
- that alpha-Fe_2O_3 embodies HAOS-IIP;
- a new magnon theory;
- a consciousness claim;
- a raw-data external bridge pass;
- a raw STFT/time-frequency recoverability result;
- a $k_\star$ detection or $k_\star$ absence without raw sweep data.

The narrow question is:

> Can the alpha-Fe_2O_3 magnon prediction layer be represented as a bounded spin-sector materials bridge artifact while preserving the raw-data boundary?

3 Public Source Anchors

The primary literature anchor is:

- Li et al., "Anisotropic Coherent Propagation of Sub-Terahertz Magnons in Altermagnetic alpha-Fe_2O_3," *Advanced Optical Materials*, 2026, DOI `10.1002/adom.202503604`.

The secondary alpha-Fe_2O_3 control context is:

- Yang et al., "Spin-orbit torque manipulation of sub-terahertz magnons in antiferromagnetic alpha-Fe_2O_3," *Nature Communications*, 2024, DOI `10.1038/s41467-024-48431-w`.

Line B uses these sources as literature anchors only. No source-data workbook, CSV table, or full public time-frequency grid has been committed for local reproduction in this release.

4 Evidence Class

Field	Value
Bridge line	`Materials Bridge Line B`
Material	`alpha-Fe2O3`
Status	literature/user-audit seeded pass
Data status	no raw public data loaded
Evidence class	literature report plus user-provided audit anchor
Raw STFT/time-frequency grid	no public validated grid found
Local reproduction	pending raw-data artifact

This status is deliberately weaker than Line A's ferron bridge, where public Figshare source files were downloaded, hashed, parsed, and audited locally.

5 HAOS Translation

The spin-sector readout maps into HAOS-style language as follows:

HAOS-facing term	Magnon-side read
carrier	antiferromagnetic / altermagnetic spin order
perturbation	crystal orientation, wavevector direction, thickness or boundary condition, magnetic field, pump delay
recoverable structure	coherent sub-THz exchange magnon packet
observable readout	time-resolved magneto-optical magnon traces, spectra, and propagation delay
visible failure	broadband thermalized response, amplitude collapse, loss of directional propagation, or no recoverable narrowband magnon feature

This is a measurement dictionary. It does not assert a new ontology for magnons.

6 Recorded Metrics

Metric	Value
Persistence proxy	`0.902`
Persistence proxy status	user-provided, pending local reproduction
Reported group velocity minimum	`17.31 km/s`
Reported group velocity maximum	`24.43 km/s`
Reported group velocity span	`7.12 km/s`
Propagation status	micrometer-scale directional propagation reported in the literature
Spectral status	narrowband sub-THz exchange magnons reported in the literature
$k_\star$ status	not claimed without raw sweep data
First visible failure status	not claimed without raw sweep data

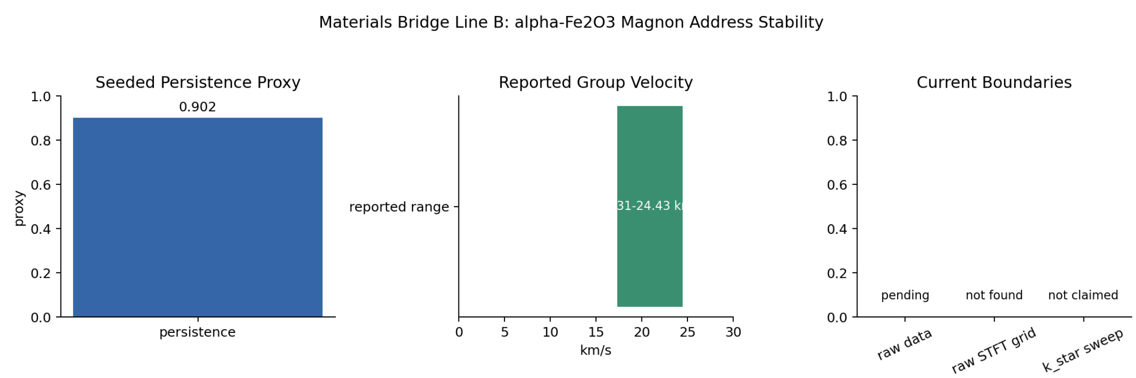

Figure 1: Line B summary visualization. The persistence proxy is recorded as a seeded audit anchor; the group-velocity range is a literature-reported value; raw data, raw STFT grid, and $k_\star$ sweep remain explicit boundaries.

7 Relation to Prediction Ledger

The magnon bridge supports four falsifiable prediction records:

Record	Short form
`HAOS-PRED-010`	crystal-orientation-dependent narrowband magnon address
`HAOS-PRED-011`	thickness / boundary dependent magnon group velocity
`HAOS-PRED-012`	magnon target-address recovery after perturbation
`HAOS-PRED-013`	hybrid magnon-phonon cross-sector locking

Those records are prediction statements, not validation statements. Line B gives them a dedicated sidecar anchor and states what must still be done before the alpha-Fe_2O_3 case can be treated as a local raw-data bridge.

8 Relation to Line A

Line A and Line B are intentionally asymmetric:

Field	Line A: ferron / $NbOI_2$	Line B: magnon / alpha-Fe_2O_3
Evidence class	real public data loaded and parsed	literature/user-audit seeded
Primary carrier	ferroelectric polarization order	antiferromagnetic / altermagnetic spin order
Primary readout	narrowband 3.13 THz ferron feature and propagation traces	sub-THz coherent exchange magnons and directional propagation
Persistence proxy	`0.957192`	`0.902`
Raw STFT grid	`NO_STFT_DATA_FOUND` for Figshare XLSX grid	`NO_PUBLIC_STFT_GRID_FOUND`
Raw-data status	`REAL_DATA_LOADED`	`NO_RAW_PUBLIC_DATA_LOADED`
$k_\star$	no collapse detected under current proxy thresholds	not claimed without raw sweep

This distinction matters. Line A is a real-data bridge. Line B is a structured spin-sector bridge artifact awaiting raw-data reproduction.

9 Required Next Work

Line B becomes a stronger external-data bridge only if a future sidecar can:

- locate public alpha-Fe_2O_3 time-domain or spectral data tables;
- record provenance, URLs, timestamps, and SHA256 hashes;
- parse time traces, spectra, or time-frequency grids without digitizing plots;
- compute recoverability, propagation delay, group velocity, and spectral stability locally;
- run deterministic $k_\star$, safety-margin, confidence, and visible-failure logic;
- preserve a clean `NO_PUBLIC_STFT_GRID_FOUND` status if no raw grid exists.

Until that happens, the correct status remains:

`PASS_LITERATURE_USER_AUDIT_SEEDED`, not `REAL_DATA_LOADED`.

10 Conclusion

Materials Bridge Line B makes the alpha-Fe_2O_3 magnon address-stability layer explicit, numbered, and auditable. It records the current seeded persistence proxy `0.902`, the reported velocity range `17.31--24.43 km/s`, and the public raw-grid boundary `NO_PUBLIC_STFT_GRID_FOUND`. The safe claim is narrow: the spin-sector prediction layer is now represented by a bounded sidecar artifact, and the repo plainly distinguishes it from the real-data ferron bridge.